INCH-POUND

A-A-59816
4 August 2008

COMMERCIAL ITEM DESCRIPTION
OPTICAL TIME DOMAIN REFLECTOMETER, SINGLE AND MULTI-MODES

The general Service Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This Commercial Item Description(CID) describes an Optical Time Domain Reflectometer (OTDR) designed to make calibrated optical attenuation and distance measurements for the purpose of diagnosing and locating faults, components, and terminations of 850 nm/multi-mode, 1300 nm/multi-mode, 1310 nm/single-mode, and 1550 nm/single-mode wavelengths using plug in laser modules. Any instrument that meets this specification will have to operate within this specified equipment environment. This CID is meant as a minimum requirement for an Optical Time Domain Reflectometer, herein referred to as an OTDR, in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. CLASSIFICATION. The instrument covered by this CID may be commercially available equipment modified to the extent necessary to meet the following specification. The equipment shall be Class 3 in accordance with MIL-PRF-28800 or its Commercial Standard equivalent, with the exceptions noted herein.

3. SALIENT CHARACTERISTICS. The instrument shall be capable of operation and product sample testing, if so requested, within the accuracies, limits, and specifications herein.

3.1 Operational requirements.

3.1.1 Safety. The instrument shall meet all safety and environmental requirements as specified in MIL-PRF-28800, or its Commercial Standard equivalent, for the classification as stated herein.

3.1.2 Dynamic Range. The instrument shall provide a minimum dynamic range of 20 dB at the largest pulse width for the following wavelengths: 85 nm/multi-mode and 1300 nm/multi-mode. The instrument shall provide a minimum dynamic range of 30 dB at the largest pulse width for the following wavelengths: 1310 nm/single-mode and 1550 nm/single-mode.

3.1.3 Attenuation Accuracy and Resolution. The instrument shall have an attenuation accuracy of at least ±3% or better for all wavelengths. The instrument shall have an attenuation resolution of at least 0.01dB for all wavelengths.

3.1.4 Maximum Measurable Distance. The instrument shall provide a useable scale reading (maximum measurable distance of the fiber) of at least the following:

Wavelength/Mode	Maximum Range
850 nm/multi-mode	15 km/ 9 miles
1300 nm/multi-mode	30 km/ 19 miles
1310 nm/single-mode	100 km/ 63 miles
1550 nm/single-mode	200 km/ 125 miles

3.1.5 Distance Resolution. The instrument shall provide a distance resolution of at least 1 meter/ 3.3 feet for all wavelengths.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/742d CBSG, 380 Richard Ray, Ste 104 Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

FSC 6625

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3.1.6 Spatial Resolution. The instrument shall provide a spatial resolution (the smallest spacing of two closely spaced reflecting elements which can be clearly resolved) of at least the following:

Wavelength/Mode	Maximum Range
850 nm/multi-mode	2 m/ 7 ft
1300 nm/multi-mode	4 m/ 13 ft
1310 nm/single-mode	8 m/ 26 ft
1550 nm/single-mode	16 m/ 52 ft

3.1.7 Reflective Dead Zone (Fresnel). The instrument shall have a maximum pulse recovery time (the amount of time required for the OTDR receiver to recover when the reflected power is large and the receiver is overdriven) of 5 meters/ 16 feet on the following wavelengths: 850 nm/multi-mode, 1300 nm/multi-mode, 1310 nm/single-mode, and 1550 nm/single-mode.

3.1.8 Non-Reflective Dead Zone (Backscatter). The instrument shall have a maximum Non-Reflective Dead Zone of 15 meters/ 49 feet on the following wavelengths: 85 nm/multi-mode, 1300 nm/multi-mode, 1310 nm/single-mode, and 1550 nm/single-mode.

3.1.9 Pulse Width. This instrument shall have the ability to have selectable pulse widths for both single and multi-modes. The instrument shall have the ability to select several pulse widths between 10 ns and 500 ns plus several between 1 µs and 10 µs. These pulse widths shall be on all of the module(s) if it uses interchangeable plug-in modules, or some other equivalent means, of accompanying this same task.

3.1.10 Real Time Monitoring. The instrument shall provide the ability to continuously update the graphical display and the alphanumeric readout. This capability shall be available for both complete trace display and expanded trace display modes of operation.

3.2 General Requirements.

3.2.1 Display, Input, and Data Storage. The instrument shall be capable of both a graphical and an alphanumeric display. Graphically the instrument shall be capable of displaying an entire trace over the maximum range and shall have the capability of displaying and expanding any portion of that over the entire range. Alphanumerically the instrument shall be capable of the attenuation measurements and distance measurements. The display shall be capable of displaying measurements in color.

The instrument shall also be capable of providing an input user's interface, as well as providing data storage capabilities internal and external storing of data.

The instrument shall be supplied with an US style English keyboard to permit easy operation.

3.2.2 Transit Case. The instrument shall be provided with a ruggedized hard transit case. Accessory storage shall be provided for in the hard transit case. It is the Government's intent that the transit case be able to withstand the normal situations found in exterior fiber optic maintenance. The transit case needs to withstand rain and dust and not crush under a 25 lb load. In Military Standards the transit case must be in accordance with paragraph 3.7 for Class 3 of MIL-PRF-28800 or its equivalent with exceptions listed below. Commercial Standards are acceptable as long as they are equivalent.

3.2.3 Splash-proof. The Government's intent is that the transit case be water resistant. This is formally accomplished by requiring splash-proof testing. The transit case shall be splash-proof to Class 1 according to MIL-PRF-28800F, paragraph 3.8.6.2. Commercial Standards may be substituted to include type of testing used to verify water resistance.

3.2.4 Dust Exposure. The Government's intent is that the transit case be dust resistant. This is formally accomplished by requiring dust exposure testing. The item and transit case shall meet Class 1 requirements as listed in MIL-PRF-28800F, paragraph 3.8.10. Commercial Standards are acceptable as long as they are equivalent.

3.2.5 Laser Design and Safety. The instrument's laser modules shall meet the safety and design requirements as applicable for the laser's safety requirements according to ANSI-Z136.1 and FDA Performance Standards for Laser Products, 21CFR1040.10.

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3.2.6 Crush Resistance. The Government's intent is that the transit case be crush resistant. The transit case must withstand a 25 lb load without damaging the case. Commercial Standards or test condition with results may be used to satisfy this requirement.

3.2.7 Connector**.** The instrument shall be equipped with a type "ST" connector and shall also be supplied with the following adapters: biconical, SMA, FC, and FC/PC. Utilizing Hybrid patchcords (fiber cables terminated with different types of connectors) is acceptable as long as they are included with the item.

3.2.8 Units**.** The instrument shall be capable of measurements using units of meters, kilometers, feet, kilo-feet, and miles.

3.2.9 I/O Ports**.** The instrument shall have as a minimum one Ethernet and one USB Input/Output Ports. Converters, such as USB/RS-232, are acceptable as long as they are included with the unit.

3.2.10 Mean Time Between Failures (MTBF)**.** The design of the instrument shall be such that under normal use and operation the instrument does not fail within 1500 hours of operation with a statistical certainty of 95%. The instrument reliability shall be in accordance with MIL-PRF-28800 paragraph 3.13.

3.2.11 Calibration and Maintenance Adjustments**.** The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 1500 hours per year.

3.2.12 Power. The instrument shall be capable of operating with the following power requirements:

3.2.12.1 AC Power**.** The instrument shall operate from nominal commercial, military, and shipboard power sources of 110 and 220 VAC (±10%) at line frequencies of 50 to 60 Hz. The appropriate US style power cord and adapter shall also be supplied with the instrument.

3.2.12.2 DC Power**:** The instrument shall be capable of operating from 10-15 VDC via an external connector (vehicle cigar style) that shall be supplied with the instrument.

3.2.12.3 Battery Power**.** The instrument shall be capable of operation from internal battery. A rechargeable battery/batteries shall be provided with the instrument. Battery life shall be at least 4 hours. A battery charger of the type appropriate for the battery/ batteries shall also be supplied with the instrument. The instrument shall be equipped with an indicator that will indicate the remaining charge and/or a low battery state. The low battery state indicator shall alert the user sufficiently in advance to allow time to shut the instrument down and not risk the failure to loose unsaved data.

3.2.12.4 Battery Compartment. The instrument's battery compartment shall meet all the specifications in paragraph 3.5.2.9 of MIL-PRF-28800 or Commercial Standard equivalent.

3.2.13 Operating Temperature**.** The operating temperature shall be 0°C to 40°C.

3.2.14 Non-Operating Temperature**.** The non-operating temperature shall be -20°C to + 60°C.

3.2.15 Operating Humidity**.** The operating humidity shall be 85% RH.

3.2.16 Physical Dimensions and Characteristics. This instrument shall not weigh more than 20 lb. This instrument shall not have a physical size larger than 18" high by 20" wide by 6" deep.

3.2.17 Operation and Maintenance Manuals**.** The contractor shall supply, at no cost to the Air Force, two operation and maintenance manuals in accordance with MIL-PRF-28800 paragraph 6.6. The operation and maintenance manual shall be written to the following levels: (1) unit operational verification and (2) module level.

4. REGULATORY REQUIREMENTS.

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5.2.1 The contracting officer has the option of accepting or rejecting the product.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 The Federal Acquisition Regulation (FAR) may be obtained at www.acqnet.gov/far or from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

7.1.3 Code of Federal Regulations standards may be obtained at http://www.gpoaccess.gov/cfr/ or from the U.S. Government Printing Office, 732 N. Capitol Street, NW, Washington DC 20401.

7.1.4 ANSI standards may be obtained at http://ansi.org/ or from the American National Standards Institute, 25 West 43 rd Street, 4 th Floor, New York, NY 10036.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Packaging requirements.

7.3 Key words.

a. digital input/output
b. output connector
c. rack-mount
d. radio frequency cable
e. time base

MILITARY INTERESTS:

Custodian:
Air Force – 99

Preparing activity:
Air Force – 84

Agent:
Air Force – 99

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